Testing the fossil maximum ages that divide land-plant molecular clocks

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Abstract

Fossil calibrations can shift molecular estimates of early land-plant diversification by tens of millions of years, largely through maximum ages: how far a lineage could predate its recognised fossil record. We reassessed the three maxima that most divide recent clocks, 515.5 and 459.1 Ma for crown land plants and 431.2 Ma for crown vascular plants, separating fossil identity and age from sampling opportunity and from the link between a diagnostic character and a crown group. The oldest maximum rests on a qualitative preservation-control argument that the available material cannot quantify. In one Libyan basin, 28 older well cuttings without trilete spores, against 51 of 59 younger cuttings with them, support a local, model-dependent limit on sustained delivery of trilete spores, chiefly in the later Ordovician; they do not establish a global absence. For vascular-plant bodies, 48 locality–age units give a conditional 95% older bound near 429 Ma on the start of their recognisable record under constant recording intensity; combining or restricting the observations moves it to 432.4–434.05 Ma. In a retrospective model with no delay before diagnostic bodies appeared, crown ages of 444–452 Ma would require earlier bodies to have been recorded about 8 to 12 times less often than later ones, or 2.4 to 4.5 times under coarser unit treatments; an assumed diagnostic delay reduces the contrast. These are compatibility requirements, not estimates of historical recording rates or delays. Admitting Ordovician sporangial fragments from Oman moves a broad body-record bound about 20 Myr older. The study identifies which observations constrain particular recording histories, and which biological assumptions are still needed to turn those constraints into molecular-clock maxima.

1. Introduction

Did the diversification that gave rise to living land plants begin in the Cambrian, or only in the Late Ordovician, tens of millions of years later? The answer changes how we read the long interval between the earliest microscopic plant remains and the appearance of branched plant bodies. An early radiation allows a prolonged history of small, local or hard-to-recognise plants; a late one places more of the diversification of living lineages close to their conspicuous fossil record. Neither crown age, by itself, dates the first occupation of land: a terrestrial stem lineage can predate the crown of living land plants, and a node date does not date the growth of substantial terrestrial biomass. The timing does bear on the order in which plant characters were assembled, and on hypotheses about early terrestrial productivity and weathering (Kenrick et al. 2012; Strother and Foster 2021; Donoghue et al. 2021).

Recent molecular clocks give markedly different answers (Fig. 1). Morris et al. (2018b) placed crown Embryophyta (the last common ancestor of living land plants and all its descendants) at 515–474 Ma across seven topologies, and Harris et al. (2022) at 515–493 Ma. Hooper et al. (2026), using the same molecular dataset as Harris et al. but recalibrated fossils, obtained 457–450 Ma, a crown-group mean 55 Myr younger. For crown Tracheophyta (living vascular plants) the estimates are 451–419 Ma (Morris et al.), 452–447 Ma (Harris et al.) and 432–428 Ma (Hooper et al.). Su et al. (2021) obtained still older crown-embryophyte ages with a Proterozoic maximum. These are disagreeing results, not one uncertain estimate.

A major source of the disagreement is how fossils set maximum ages. A fossil can establish that a lineage existed by a particular time, provided its age and placement are sound. Arguing that the lineage could not be much older needs more: a reason to think that older rocks would probably have preserved, received and yielded recognisable evidence had the lineage existed. That reason involves both the fossil record and the biology of the earliest members of the lineage. Morris et al. set the crown land-plant maximum at 515.5 Ma; Hooper et al. set it at 459.1 Ma from the oldest possible trilete spores, and set 431.2 Ma for crown vascular plants (Table 1). The use of absence-based maxima is itself contested (Hedges et al. 2018; Morris et al. 2018a; Su et al. 2021).

Three kinds of observation bear on these maxima: a dispersed reproductive character (a spore with a particular wall or mark), associated sporangial or body material, and a fossil diagnostic enough for crown placement. They are not successive grades of more advanced plants. Each has its own phylogenetic implications and its own route into the sampled record. Linking any of them to a node crosses two gaps that Marshall (2019) separates: from divergence to the first diagnosable character, and from that character to its first fossil. Early land plants strain both. Dispersed land-plant spores appear long before branched bodies (Wellman and Gray 2000), terrestrial deposits are scarce before the latest Silurian, at least in western Europe (Kenrick et al.

2012), and the oldest parent plants known in detail are minute (Dianne Edwards et al. 2014).

Table 1. The three maxima examined, what each rests on, and what remains conditional.

Maximum	Set by	Evidence it rests on	What the record lets us test	What stays conditional
515.5 Ma, crown land plants	Morris et al.	Base of the *Olenellus* Biozone, below the oldest possible Cambrian cryptospore-like forms; older marine sequences with algal cysts but no land-plant spores	The affinity of the Stage 4 material; whether any enumerated pre-515.5 Ma sequence exists	Spore delivery to marine basins, catchment, search sensitivity, and spore production by the earliest crown members
459.1 Ma, crown land plants	Hooper et al.	GTS2012 Sandbian base plus uncertainty, keyed to the oldest possible trilete spore (Borenshult-1, Sweden)	One local comparison of older non-detections and younger positives (Murzuq Basin, Libya)	Crown status of trilete producers, persistence of the trait, presence in the catchment
431.2 Ma, crown vascular plants	Hooper et al.	GTS2012 Homerian base plus uncertainty, keyed to Irish Homerian *Cooksonia*	A conditional bound from the positive body record, and what older histories would require	Recording before the oldest find, delay before diagnostic bodies, crown membership of the anchor

We asked what each maximum requires of the fossil evidence, which of those requirements published observations can test, and how the numerical conclusions change when recording intensity, dating or observational independence changes. We combine a source-based review of the calibration fossils, a local comparison of spore non-detections with positive records, and conditional bounds from the early body-fossil record. The general principles involved are established: incomplete recovery makes first appearances younger than the histories they record, and maximum calibrations need assumptions beyond the fossils themselves (Marshall 2019). What we add is their application, against specific source records, to the maxima that currently divide the clocks, with the size of each required assumption stated.

The body-record analysis needs no inventory of searched but empty rock. It substitutes an assumption about the rate of recorded, identifiable occurrences, before and within the record, and we vary that assumption explicitly. The rate combines how common and widespread the producers were, whether they had recognisable characters, delivery and preservation, and collection and reporting; these are not separated here.

2. Evidence

Three bodies of evidence carry the tests (Fig. 2).

Murzuq Basin cuttings. Abuhmida (2013) reported palynology from Ordovician to lower Silurian wells in southwest Libya. We transcribed all 164 appendix samples and their chart and prose reports, and used the chart-coded cuttings of five wells as the primary frame. Three wells supply 28 cuttings older than the Llandovery, none reporting trilete spores, and all five supply 59 Llandovery reference cuttings, 51 of which report them. Of the older cuttings, 10 are broadly dated Darriwilian samples from two wells and 18 are Katian or Hirnantian, wholly younger than 459.1 Ma (Fig. S2). Northern Africa lay near the Ordovician

South Pole, within the area of Hirnantian glaciation (Cocks and Torsvik 2021; Scotese et al. 2021), so this frame samples one high-latitude basin.

QSIM-801. Vecoli et al. (2017) tabulate 99 cryptospore occurrences in 17 of 20 samples from a Saudi Arabian borehole assigned to the late Dapingian–early Darriwilian, and report no trilete spores in the studied sequence. The material and basin are independent of Murzuq.

Dated body fossils. We assembled the accepted land-plant body occurrences from 453 to 393 Ma that could be dated and linked to a locality: Paleobiology Database (PBDB) names, classified by AI with cited sources and audited, together with our curated set of the oldest records and their competing readings. The counting unit is a locality–age proxy, so several collections, taxa or papers from one horizon count once. Three nested categories are declared. Land-plant bodies are any accepted macro- or mesofossil, sterile axes included; polysporangiophytes need documented branched axes bearing sporangia, or crown-member characters; crown vascular-plant members belong to the lycophyte line or the euphyllophytes. Tracheids alone do not admit a unit to the polysporangiophytes, and a record not assessed for branching is not treated as lacking it (Supplement S5.1). The main reading yields 78 land-plant body units, including the spore-filled sporangial fragments from the Ghaba-1 borehole, Oman (Wellman, Osterloff, and Mohiuddin 2003), as one Katian unit; 71 polysporangiophyte units; and 50 crown vascular-plant units. Of these, 76, 69 and 48 are certainly older than the end of the Emsian (393.47 Ma).

The body record is geographically concentrated where it matters most. Of the 48 crown vascular-plant units, only 16 have age support older than the Silurian–Devonian boundary, and those come from seven regions. The two oldest-possible clusters are Yea, Victoria, two horizons whose age is disputed between Gorstian and Early Devonian (Earp 2024), and the Ludfordian of Bathurst Island, Arctic Canada (Kotyk et al. 2002). Silurian land-plant bodies more generally come mainly from Laurentia (11 of 34 wholly Silurian units) and peri-Gondwanan Europe (8). Laurentia supplies 29 of the 78 land-plant units, most of them Devonian.

The three comparisons are independent of one another in material. Within each, independence is limited: Murzuq's 87 cuttings come from five wells and its older cuttings from three, and body units share regions and collectors, with a proxy for a horizon that is not a verified bed. None is independent of the clocks in the way a validation would need. The oldest *Cooksonia* and Ludlow *Zosterophyllum* are themselves calibrations, and the oldest trilete spores set a maximum. Agreement between a body-record bound and a clock therefore tests whether an absence-based maximum is reasonable; it does not validate a clock posterior or rank one clock above another.

Table 2 sets out the records on which the conclusions turn.

Table 2. Pivotal records: what each shows directly, how we use it, and what a crown claim would additionally need.

Record	Observed or documented	Age support	Our use	A crown claim would also need
Ghaba-1 fragments, Oman (Wellman, Osterloff, and Mohiuddin 2003)	Sporangial fragments with in situ cryptospores, some with a sporangial covering; no preserved associated axes or demonstrated conducting tissue	Katian on later correlation (Caradoc in the original)	Land-plant body, one unit	Characters placing the producer in a crown group; not a polysporangio-phyte or crown member

Record	Observed or documented	Age support	Our use	A crown claim would also need
Borenshult-1 triletes, Sweden (Rubinstein and Vajda 2019)	Possible trilete at 48.2 m; definite triletes from 43.1 m; the figures disagree at 41.1 m	Sandbian; the oldest definite level lies below the 454.41 Ma bentonite	The 459.1 Ma maximum's basis; a weak absence screen	An argument placing the producer in the crown; a trilete mark alone does not
Cooksonia barrandei, Czech Republic (Libertín et al. 2018)	Branched fertile sporophyte with terminal sporangia and in situ trilete spores; no tracheids reported	Sheinwoodian, horizon inferred from slab fauna and correlation	Oldest polysporangiophyte	Placement on the lycophyte or euphyllophyte side; tracheids alone would not suffice
Irish Homerian axes (D. Edwards and Feehan 1980)	Fragmentary sporangia on short axes, one branched; no tracheids seen	Late Homerian graptolite zone	Hooper et al.'s 431.2 Ma anchor	As for *C. barrandei*
Yea *Baragwanathia*, Victoria	Microphyllous leafy axes; no reproductive structures reported from the material claimed as Silurian (Libertín, Kvaček, and Bek 2024)	Gorstian on graptolites versus Early Devonian on brachiopods and mapping (Earp 2024); full range carried	Influential crown vascular-plant unit	Matching the competing dates to the exact horizons
Bathurst Island zosterophylls, Arctic Canada (Kotyk et al. 2002)	Fertile zosterophylls in spikes and clusters	Late Ludlow on conodonts, graptolites and brachiopods; possibly Pridoli	Influential crown vascular-plant unit (lycophyte line)	None for crown Tracheophyta; not crown Lycophyta

3. Results

3.1 The three maxima rest on different kinds of evidence

515.5 Ma: affinity and a qualitative preservation control. The maximum of Morris et al. (2018b) is placed below the oldest possible total-group records: cryptospore-like packets and polyads from the Cambrian Stage 4 Rome Formation, so the base of the *Olenellus* Biozone, about 515.5 Ma. Its absence argument is that land-plant spores share the preservation potential of the marine algal cysts with which they usually occur, so that older sequences with such cysts but no land-plant spores record a real absence (Morris et al. 2018b, 2018a). On affinity, none of the three sources we examined accepts the Stage 4 material as land-plant spores: Strother (2016) placed the Tennessee taxa in Charophyta, Taylor and Strother (2024) read the packets as pre-embryophytic, and Wang, Xu, and Yin (2022) favoured ancestral aeroterrestrial algae. That the material is younger than the maximum (514.5–506.5 Ma) is by design. On absence, the control shows that sporopollenin-walled microfossils survived and were recovered. It does not by itself show that terrestrial spores reached those marine basins at comparable rates, that the basins drew on the land areas where early plants grew, that preparation and search would have found small cryptospores, or that the earliest producers made recognisable dispersed spores. No source we assessed supplies an enumerated, comparably searched pre-515.5 Ma sequence with sample-level counts, so the argument cannot be given a calibrated non-detection probability here.

459.1 Ma: a stage-base rule. Hooper et al. (2026) key this maximum to a queried *Chelinospora?* specimen at 48.2 m in the Borenshult-1 core, southern Sweden (Rubinstein and Vajda 2019), and take the number from

the stage, not the level: the GTS2012 base of the Sandbian plus its uncertainty, 458.4 + 0.7 = 459.1 Ma. The describing authors plot that specimen as a possible trilete only. Their oldest definite trilete level, at 43.1 m, is also Sandbian and lies below the 454.41 Ma Kinnekulle K-bentonite (Supplement S2.4). The stage-base rule gives the same maximum from either level, so the value does not rest on the doubtful specimen. Accepting the definite triletes could move a minimum, not this maximum, and only with a further argument that their producers were crown members. As an absence test the core is weak: its three Darriwilian samples below the oldest definite trilete cannot exclude reference-level intensity under any of the three readings of its figures ().

431.2 Ma: an anchor and an older candidate. Hooper et al. anchor the vascular-plant maximum on the Irish Homerian *Cooksonia*, again by the stage-base-plus-uncertainty rule: 430.5 + 0.7 = 431.2 Ma on GTS2012. Their calibration table does not discuss the older *C. barrandei* (Table 2), which carries the same characters, a branched sporophyte with terminal sporangia and no demonstrated tracheids, from a Sheinwoodian horizon inferred from slab fauna and correlation. If the rule is retained, that record should be considered alongside the Irish material. On the current timescale the rule gives 430.6 + 1.3 = 431.9 Ma for the Homerian and 432.9 + 1.2 = 434.1 Ma for the Sheinwoodian, about 2.2 Myr apart; the Sheinwoodian value is about 2.9 Myr older than the published 431.2 Ma, part of which is the change of scale. An older stem-group fossil does not contradict a younger crown maximum, and neither number dates the fertile slab, whose horizon is inferred.

3.2 One Libyan basin constrains a local history of trilete-spore delivery

Hooper et al.'s ancestral-state reconstructions make a proximal trilete aperture modal at the embryophyte crown (Supplement S6). A crown older than 459.1 Ma would therefore imply undetected trilete producers somewhere, and undetected local producers only if the state persisted in early lineages, they made recognisable trilete spores, and some lived in the sampled catchment. Murzuq is the only frame we found in which that implication can be confronted with documented older and younger samples. Its provenance is mixed. For part of the sequence the source states the absence of trilete spores explicitly; for the Hawaz cuttings it lists cryptospore assemblages without an explicit trilete-absence statement, and their zeros are operational, resting on complete reporting charts and the stated microscopy protocol (Abuhmida 2013). Both support a declared observation rule; neither is a database gap.

We modelled each cutting's chance of reporting triletes as an encounter process sharing a well effect, with older encounter intensity expressed as a fraction f of the Llandovery reference. An assumed delivery-and-recovery ratio ρ covers differences in transport, preservation and preparation between older and younger material. We call a procedure *qualified* when its simulated coverage reached 0.93 on a declared grid of generating scenarios (Supplement S2); this checks the method on that grid, not the historical sampling process.

Table 3. What the Murzuq comparison allows: model-conditional 95% compatibility limits, assessed on the declared simulation grid, on older trilete-producer intensity as a fraction f of the Llandovery reference, primary stage allocation. Rows 1–2 constrain intensity sustained from 459.1 Ma to the reference; rows 3–4 constrain an interval of fixed intensity beginning at 459.1 Ma and extending further back.

Question asked of the 28 older cuttings	Delivery equal ($\rho = 1$)	Delivery halved	Delivery quartered	Delivery a tenth
Largest constant producer intensity, 459.1 Ma to the Llandovery	0.091	0.182	0.364	0.910
Largest intensity, exposure-weighted history class	0.184	0.258	0.421	0.921
Oldest end of a reference-intensity interval beginning at 459.1 Ma	462.6 Ma	464.1 Ma	467.1 Ma	no limit
Oldest end of an interval at 0.1 or 0.01 of reference intensity	no limit	no limit	no limit	no limit

What the zeros constrain is delivered, recovered encounter, sustained over the later Ordovician in one basin, since most exposure comes from the 18 younger cuttings (Table 3). Under the constant history the product

ρf stays at 0.091 across delivery scenarios; under the exposure-weighted history it runs from 0.184 at equal delivery to 0.092 at a tenth. If older material delivered and preserved spores as well as the Llandovery reference, trilete producers at constant intensity from 459.1 Ma would have had to be, under the model, below a tenth of their Llandovery intensity, and below about 0.18 of it under the exposure-weighted history. If delivery was a tenth as good, producers at nearly the reference intensity are allowed. Intervals of producers at a tenth or a hundredth of reference intensity are unbounded under every delivery scenario. The record therefore excludes, under the model, abundant, well-delivered trilete producers sustained through the later Ordovician in this basin, not sparse or poorly delivered ones, and it cannot tell those apart from a lineage that lived outside the basin's catchment. A local absence of delivered spores is not a global absence of producers.

The numerical strength of this inference depends most on how the cuttings are assumed to depend on one another. The primary model treats cutting outcomes as independent given a shared well effect. Fitting a separate rate to each well barely moves the constant-history comparison contour (0.0995 against 0.0910). Weighting each cutting's likelihood by the inverse size of its well–formation or well–stage block, so that each block carries unit total weight, gives contours of 0.393 and 0.419. These are unqualified composite-likelihood sensitivities, not a fitted dependence model, and the data do not identify a preferred dependence structure; the conclusion that matters is that "a tenth" leans on conditional independence within wells (Supplement S2.5). The other auxiliary assumptions matter too. Removing one well at a time spans exposure-weighted limits of 0.185–0.365. The pre-maximum window rests on the ten Darriwilian cuttings, and how their broad stage ages are turned into exposure decides it: shared midpoint ages give the 459.1–462.6 Ma window no exposure at all. With no contaminant transfer to older cuttings, a reduction of more than about 4.4% in the indigenous share of reference intensity crosses the short window's comparison threshold. At that window's simulated boundary, 20 of 1,000 screen-passing simulations missed the truth, and those 20 were among only 60 that made a claim, so a finite boundary claim from this frame should not be read as an unrestricted 95% guarantee.

QSIM-801 points the same way qualitatively: its authors report no triletes in a cryptospore-rich Middle Ordovician sequence. Treating each of the 99 occurrences as an independent unit on which a trilete would have been detected, a trilete-to-cryptospore ratio that stayed above 0.031 in every sample is incompatible with the record at equal effective recovery, and one above 0.307 at a tenth of it (as shares of all spores, 0.030 and 0.235). The source does not document the search protocol needed to make those units independent and target-detectable, so this is an exploratory, model-conditional calculation, not a second qualified test (Supplement S3).

3.3 Body fossils set conditional requirements for older histories

No assessed source supplies a roster of searched Silurian beds with comparable effort for the 431.2 Ma maximum. The Irish late-Wenlock sections report fossil-free mudstone qualitatively, without an enumerated roster (D. Edwards and Feehan 1980). We therefore turned to the positive record.

Method. Let $w(t)$ be a fixed relative rate of recorded, identifiable occurrences of a fossil category. For n independent units older than a fixed younger cutoff c, the one-sided 95% older bound T on the start of the recognisable record solves $U(T) = U(A)\,0.05^{-1/n}$, where A is the oldest unit's age and $U(a) = \int_c^a w(s)\,ds$. With constant w this is the classical confidence interval on a stratigraphic range (Strauss and Sadler 1989; Marshall 1990), and the absolute rate cancels. Interval-dated units enter conservatively: the oldest possible age and the smallest number certainly older than the cutoff. The bound is a confidence statement under a fixed observation model and the adopted numerical timescale, not a posterior probability for the crown. We report six histories: constant intensity; intensity proportional to PBDB land and marginal-marine collections per Myr, with and without collections that match the plant-filtered extract; and intensity declining exponentially into the past on 25 and 10 Myr timescales, the last also combined with plant-excluded collection density (Supplement S5).

The crown vascular-plant bound. For the 48 crown-member units the bounds are close together under the histories tied to the observed record: 428.8 Ma for constant intensity, 428.9 Ma for collection density and 428.5 Ma with plant-matched collections removed (Table 4). A 25 Myr decline into the past gives 431.6 Ma, and a 10 Myr decline gives no bound within the 469.4 Ma limit of the effort history. The unit definition matters as much. Under constant intensity, collapsing units to region–stage cells (19 units) gives 432.4 Ma

and the curated oldest-record set alone (15 units) gives 434.05 Ma, both older than 431.2 Ma. These are not estimates of an effective sample size, and the curated set also changes what was searched, but they show that plausible ways of organising the same records move the bound across the maximum. Other declared sensitivities stay close to the primary value: individual collections 428.0 Ma, wide-age records restored 428.1 Ma, each region left out in turn 427.2–430.5 Ma, and a window ending at the Silurian–Devonian boundary (13 units) 428.5 Ma.

Shifting every stage boundary together by its quoted uncertainty, or alternately in opposite directions, moves the constant-intensity bound between 427.31 and 430.42 Ma, by at most 1.58 Myr. It stays younger than the maximum in each tested scenario, whether 431.2 Ma is held fixed, moved with the shifted stages, or recomputed by the stage-base-plus-uncertainty rule; under the 25 Myr decline it crosses. These are deterministic scenarios, not a probability interval. Removing the three Yea units changes the constant bound by −1.67 Myr, and removing the five Bathurst Island units by 0.26 Myr (positive is older), because the count falls while Yea stays oldest. Removing *C. barrandei* changes the polysporangiophyte bound by −2.38 Myr (Supplement S5.3). Simulations at the real unit counts show that the calculation behaves as stated when its assumptions hold (exact-age coverage 0.941–0.958) and fails when they do not: duplicated horizons treated as independent cut coverage to about 0.78 (pairs) or 0.53 (quartets), and fitting constant intensity to a true 10 Myr decline gives 0.03–0.24 (Supplement S5.6).

Table 4. Conservative 95% older bounds (Ma) on the start of the recognisable record of each body category, main reading, conditional on each recording history, the unit rule and independent units. The land-plant body column includes the Katian Oman sporangial fragments; the next column omits them. † One illustrative scenario chosen after the main analysis; not fitted, not a threshold and not a nominal 95% bound. "None": no bound within the 469.4 Ma limit of the effort history.

Recording history or unit rule	Land-plant body	Land-plant body, without Oman	Polysporangiophyte	Crown vascular plant
Constant	455.2	434.5	434.6	428.8
Proportional to PBDB collection density	456.6	436.4	436.7	428.9
Collection density, plant-matched collections removed	456.1	435.7	436.0	428.5
Declining into the past, 25 Myr timescale	465.2	437.2	437.6	431.6
Declining into the past, 10 Myr timescale	none	none	none	none
Constant, region–stage units	—	—	—	432.4
Constant, curated oldest-record set only	—	—	—	434.05
Constant, earlier intensity reduced tenfold before 426.7 Ma†	—	446.7	447.9	448.1
Constant, earlier intensity reduced a hundredfold before 426.7 Ma†	—	none	none	none

What an older crown would require. The sequence says nothing about the rate before the oldest possible crown-member find, so an older crown age can be accommodated by lower recording intensity before that find, by a delay before diagnostic bodies appeared, or by both. We asked how large each would have to be (Fig. 4). With a step in the rate at the oldest possible find (426.7 Ma) and no assumed delay, so that the recognisable record starts at the candidate lineage age, 431.2 Ma would need earlier records at 0.48 of the later rate, and 444, 447 and 452 Ma would need 0.124, 0.105 and 0.085 of it. Under the coarser region–stage or curated units, 431.2 Ma needs no reduction at all and 444–452 Ma needs 0.22 to 0.42 of the later rate. These are retrospective compatibility contours: the step sits at an age chosen after seeing the data, the 0.05 rule gives them no nominal confidence, and they are not fitted rates or estimates that early plants fossilised at a tenth of the later rate. The tenfold scenario in Table 4 is one point near this frontier. The fitted records look the same under every point on it; the answer is decided in an interval the body record cannot see.

A recognisable-record endpoint is not a crown date. Mapping one to the other needs an assumed delay be-

tween crown origin and the recognisable presence of crown-diagnostic bodies (Marshall 2019). With constant intensity, reaching the young end of Harris et al.'s interval (447 Ma) needs an assumed delay of at least about 18 Myr at the 95% endpoint bound; a 20 Myr delay maps 428.8 Ma to 448.8 Ma. These are minimum offsets, not fitted delays. Delay and earlier recording rate trade off: at a given candidate crown age, an assumed delay reduces the required recording contrast (Fig. 4; Supplement Table S9b). An ignored 10–20 Myr origin-to-body delay can give zero coverage of the lineage age (Supplement Table S11).

3.4 Fossil category and recovery route move the bound

Including the Ghaba-1 fragments as one land-plant body unit, with oldest possible age 452.8 Ma, gives bounds of 455.2–456.6 Ma under constant and effort-tracking intensity and 465.2 Ma under the 25 Myr decline. Without them the oldest unit is the Sheinwoodian *C. barrandei* and the bounds are 434.5–437.2 Ma (Fig. 3, top). One unit moves the bound by about 20 Myr. With it, every declared history fits the sequence poorly (constant intensity: nominal $p = 5.5 \times 10^{-12}$ at midpoint ages), because the Katian unit is followed by more than 12 Myr without another, and no assignment of ages within the recorded ranges rescues any history (§3.5). These bounds are therefore sensitivity arithmetic, not a date at which land-plant bodies became recognisable. Comparing them with the 458.2 Ma minimum for land-plant cryptospores does not estimate a delay between origin and bodies in either direction: that value is a fossil minimum, not an origin.

One plausible contributor to the poor fit is the pooling of records recovered by different routes. The Oman fragments came from palynological preparation of borehole material (Wellman, Osterloff, and Mohiuddin 2003), whereas most later units are compressions collected from bedding planes, as at Capel Horeb, where collections came from a quarry face and its spoil heaps (D. Edwards and Rogerson 1979). Targeted microscopic searches for producer fragments and the collection of larger compressions do not offer equivalent recording opportunities. The present analysis does not identify the cause of the misfit or estimate an origin-to-body delay. The same dependence on the oldest admitted records applies to the crown vascular-plant bound, which rests on Yea and Bathurst Island.

3.5 Which recording histories conflict with the observed sequence

The shape diagnostics compare the observed age sequence with each history by a Kolmogorov–Smirnov statistic conditional on the count and the maximum, using ages allocated within their reported ranges. Under the allocations drawn, a steep (10 Myr) decline fits the crown-member sequence poorly, and constant intensity fits under independent allocation ($p = 0.056$–0.715, 5th–95th percentiles) but not at midpoint ages ($p = 0.0085$; Supplement Table S7).

A history that fails under the allocations drawn may fit another that the ranges admit, so we searched all admissible allocations exhaustively for the one most favourable to each history, including whether each of the two units that straddle the Emsian cutoff falls inside the window (Supplement S5.2). Under the finite-sample reference, the 10 Myr decline fails in every inclusion state: no admissible age allocation reaches $p = 0.05$. The 25 Myr decline has a passing allocation with all 50 possible units ($p = 0.067$); with one straddling unit it falls just short ($p = 0.049$), and with the 48 guaranteed units it fails ($p = 0.035$). Plant-excluded collection density combined with the 10 Myr decline passes when one or both straddling units are counted. For land-plant bodies with the Oman unit, none of the six histories has a passing allocation in any inclusion state.

A failure across every admissible allocation is a conservative rejection of that fixed history for this diagnostic under an observation model in which the recorded supports contain the true ages and inclusion state almost surely: the largest finite-sample p over those supports cannot be smaller than the p at the true ages. This fixes the history, cutoff, timescale, inventory and unit rule and assumes independent continuous occurrence ages. If containment can fail, its error probability must be added to the nominal test level; geological ranges have no demonstrated joint coverage. The supports are encoded as independent rectangles, so a coherent section-level chronology could exclude some passing allocations but cannot rescue a failure. A passing allocation shows only that the history is possible under the diagnostic; it is not a fitted chronology, evidence of model adequacy or a ranking of histories. So the steep decline conflicts with the crown-member sequence, the 25 Myr decline does not when every possible unit is counted, and constant intensity is not shown to be adequate.

4. Discussion

4.1 An older, sparsely recorded flora is allowed, not required

Where the record can be interrogated, it places conditional limits on two specific observation histories: sustained, well-delivered Llandovery-level trilete encounter through the later Ordovician in the Murzuq Basin, and the start of the recognisable crown vascular-plant body record under the later Silurian–Devonian recording rate and primary unit rule (one-sided 95% older bound near 429 Ma). Neither limits early land plants that were rare, local, small, poorly preserved or seldom recognised. Murzuq allows trilete producers at a tenth of their later intensity, at full intensity with poor delivery, or living outside the basin's catchment. The body record allows crown vascular plants that were present but recorded several times less often before the Ludlow, or that lacked crown-diagnostic bodies.

The record cannot say whether such histories happened. Several processes could produce them: producers that were less common or more geographically restricted; growth in settings with little contemporaneous terrestrial deposition, which is scarce before the latest Silurian (Kenrick et al. 2012); smaller or less diagnostic bodies, since size, cuticle and lignified tissue raise the chance that a body is preserved, found and recognised, and those traits evolved along the lineages being dated; and less searching of the relevant intervals. The oldest branched land-plant sporophyte is preserved in marine sediments (Libertín et al. 2018), and the Ordovician Oman fragments show that small sporangia without preserved axes can survive. We do not prefer a scenario: a statement that a given contrast or delay is modest or likely would need a separate biological argument. A lower rate of recorded occurrences is also not the same as a lower probability that an individual plant fossilised.

4.2 Relation to earlier fossil-based estimates

The closest empirical precedent is Silvestro et al. (2015). They modelled genus-level PBDB plant occurrences with a joint Bayesian model of preservation and diversification and estimated the origin of vascular plants at a mean of 433.5 Ma (95% highest posterior density 449.0–424.0 Ma), offering the posterior as a calibration density. Their target, units and treatment of recording differ from ours. They estimate when the earliest vascular-plant genus lineage originated, from a macrofossil database with a fitted, heterogeneous preservation rate; we bound the start of the recognisable record of a declared crown-member category, from a reviewed locality–age inventory, under stated fixed recording histories, and then ask what older crown ages would require. Overlap between their posterior interval and our conditional bounds is neither validation nor contradiction: they are different objects with partly shared data. Methods such as CladeDate generate calibration densities directly from fossil times (Claramunt 2022); we report bounds and compatibility requirements instead, because the category definitions, unit rules and recording history before the oldest find are where the present dispute lies. Hooper et al. already link spore-character reconstruction to revised calibrations; our contribution is to examine the observation and recording assumptions that connect those interpretations to maxima.

What the clocks need from this is an account of assumptions, not a ranking. Under constant or effort-tracking intensity and the primary unit rule, the crown vascular-plant bound lies inside Hooper et al.'s 432–428 Ma interval and inside the full Morris et al. range (451–419 Ma). The Morris et al. means near 444 Ma and Harris et al.'s interval would need earlier crown vascular-plant bodies to have been recorded less often than later (about 8 to 12 times under the retrospective step model, primary units and zero delay; 2.4 to 4.5 times under coarser units), or crown-diagnostic bodies that arose at least about 15–23 Myr after the node (minimum offsets at the constant-intensity bound), or some combination. Agreement between the constant-intensity bound and the youngest clock is not independent support for that clock: nothing shows constant intensity is biologically appropriate near a clade's origin, and the same fossils are among its calibrations. Soft-bound maxima are a legitimate way to express this uncertainty, provided the mass and shape of the tail are justified against an explicit recording history and character history.

4.3 Which conclusions depend on which assumptions

Table 5. Dependence of each main conclusion on recording assumptions and on the character-to-lineage mapping.

Conclusion	Rests on	Fails if
Murzuq excludes abundant, well-delivered trilete producers sustained through the later Ordovician	encounter model; delivery ρ; reference purity; stage dating of cuttings, chiefly 18 later-Ordovician ones; cutting outcomes independent given a shared well effect	older delivery was much lower, reference triletes include caved or recycled grains, ages are shifted, or cuttings within a well and formation are strongly dependent (an unqualified block-weighting sensitivity gives a comparison contour near 0.4)
That exclusion bears on crown land plants	producers lived in the catchment; trilete production at the embryophyte crown and its persistence (Hooper et al. 2026); our persistence pilot	early producers lived elsewhere, did not make recognisable trilete monads, or lost them in lineages present in the basin
The one-sided 95% older bound on the start of the recognisable crown vascular-plant body record is about 429 Ma	recording history before and within the record; unit rule and independence; ages within reported ranges; adopted stage boundaries	earlier recording was several times lower (§3.3); units are pseudoreplicated or coarser (region–stage or curated units give 432.4–434.05 Ma); Yea or Bathurst ages are wrong
That bound bears on the crown vascular-plant node	lycophyte-line and euphyllophyte bodies are crown members; crown-diagnostic bodies arose near the node	crown origin preceded recognisable crown bodies by a long delay
The 10 Myr decline conflicts with the crown-member sequence	supports contain the true ages; finite-sample reference; independent units	supports are too narrow, or units are dependent
The land-plant body bound is 455.2 Ma under constant intensity	the Oman fragments are one unit of the declared category; recording history	the fragments are excluded (434.5 Ma) or split; the poorly fitting histories misdescribe the sequence

The right-hand columns separate assumptions about how fossils were recorded from assumptions about what the fossils were. More searching of empty rock cannot settle the second kind: no number of barren samples shows that a body with sporangia but no demonstrated tracheids belongs to the vascular crown, or that a trilete mark means an embryophyte producer. Better-preserved specimens can, by adding characters. Under stochastic-mapping models fitted to Hooper et al.'s deposited data, a trilete aperture present at the land-plant crown persists to the vascular-plant crown with probability 0.977, conditional on the character coding, model, tree and chronology; it is not the share of extinct local producers that made recognisable trilete spores. Other links, notably from the oldest branched sporophytes to vascular crown membership, remain contested (Supplement S6, Fig. S7).

4.4 What would sharpen the answer

Four kinds of evidence would most improve these inferences. Any author could use them, and none is requested here.

1. Documented searched-but-empty Silurian non-marine beds with stated effort, which would turn the body record's unobservable early recording rate into an estimable quantity.
2. A second older-empty, younger-positive spore sequence outside polar Gondwana, with counted target and control grains, and an enumerated pre-515.5 Ma marine sequence with land-plant and algal counts.
3. Finer, independent dating of the Yea and Bathurst Island horizons, which set the crown-member bound.
4. Characters placing the oldest polysporangiate bodies on the lycophyte or euphyllophyte side of the vascular crown; tracheids alone would improve their placement but would not settle it.

5. Conclusions

Published early plant fossils place useful constraints on particular histories of spore delivery and body recording. They constrain crown ages more weakly when the earliest producers' distribution, recording intensity and diagnostic characters are left unspecified. The three maxima examined here therefore need different justifications: a qualitative preservation control at 515.5 Ma; a geographically limited non-detection comparison at 459.1 Ma, whose strength depends on how the Libyan cuttings depend on one another; and, at 431.2 Ma, an assumed link from the positive body record to crown origin. Under constant recording and the primary unit rule, the body-record bound is younger than that maximum. Older crown ages remain compatible if identifiable bodies were recorded several times less often before the oldest find, or appeared some time after the node. One Ordovician fragment assemblage moves a body-record bound by about 20 Myr, so such bounds are set by the fossils admitted to them. Future clock analyses can use this evidence by declaring the character, recording and dependence assumptions behind each maximum, and by examining how alternative defensible assumptions change the resulting timescale.

Methods in brief

All ages use the ICS v2024/12 stage bases held in the project timescale. Literature dated to GTS2012 was converted at stage level, with source-specific statements retained. Each analysis declares a named reading of disputed identifications (main or inclusive). An explicit non-detection is scoped to the material examined; a missing report becomes a zero only after a full reporting audit. The Murzuq encounter model, its simulation-based qualification and the contamination identification result are in Supplement S2. The QSIM floor calculation is in S3 and the Cambrian affinity review in S4. The Borenshult-1 calibration material is in S2.4. The body-record inventory, unit rules, age envelope, recording histories, shape diagnostics, interval search, Oman sensitivity, frontier and simulations are in S5, and the specimen-to-node pilot in S6. Supplement S7 records analyses superseded during review.

Roles of AI systems and humans

Barry W. Brook set the programme and its direction, and retains decisions on external correspondence, release and submission. This is a draft awaiting his final review. Claude Opus 5.5 (Anthropic, `claude-opus-5-5`, in Claude Code) designed the absence test and its engine, proposed the QSIM and body-record analyses, reviewed the sources, drew the figures and wrote the paper, supplement and brief. OpenAI Codex (GPT-6; from 7 November reporting `gpt-6.1-sol` as its parent model) implemented the Murzuq observation audit, the reviewed body-record analysis and its sensitivity, interval-feasibility, frontier, timescale and dependence analyses, and the public reproduction and claim-binding tools. `gpt-6.1-sol` with web search classified the 241 PBDB plant names with cited sources. Each agent reviewed the other's work; agreement between them is not independent validation. GPT-6 Pro (OpenAI) reviewed the repository twice at Barry W. Brook's request, at commits `60c406b` and `342153a`, reproducing the central calculations and identifying errors that this version corrects. Who did what, and when, is recorded in `docs/charter.md`, `docs/decisions.md` and `docs/audits/`.

Data and code

The repository holds the source-linked derived tables, model code, frozen qualification evidence, reviewed body-record outputs with hash receipts, figure and table scripts, and the audit trail. Copyrighted source texts and third-party deposits are not redistributed; scripts fetch the deposits and verify their hashes. The frozen PBDB all-taxa collections extract that the effort histories use is archived under CC0 at https://doi.org/10.5281/zenodo.24028289, so the derived tables can be rebuilt from raw input (Supplement S8). Public release of the paper is Barry W. Brook's decision.

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501 **Fig. 1.**

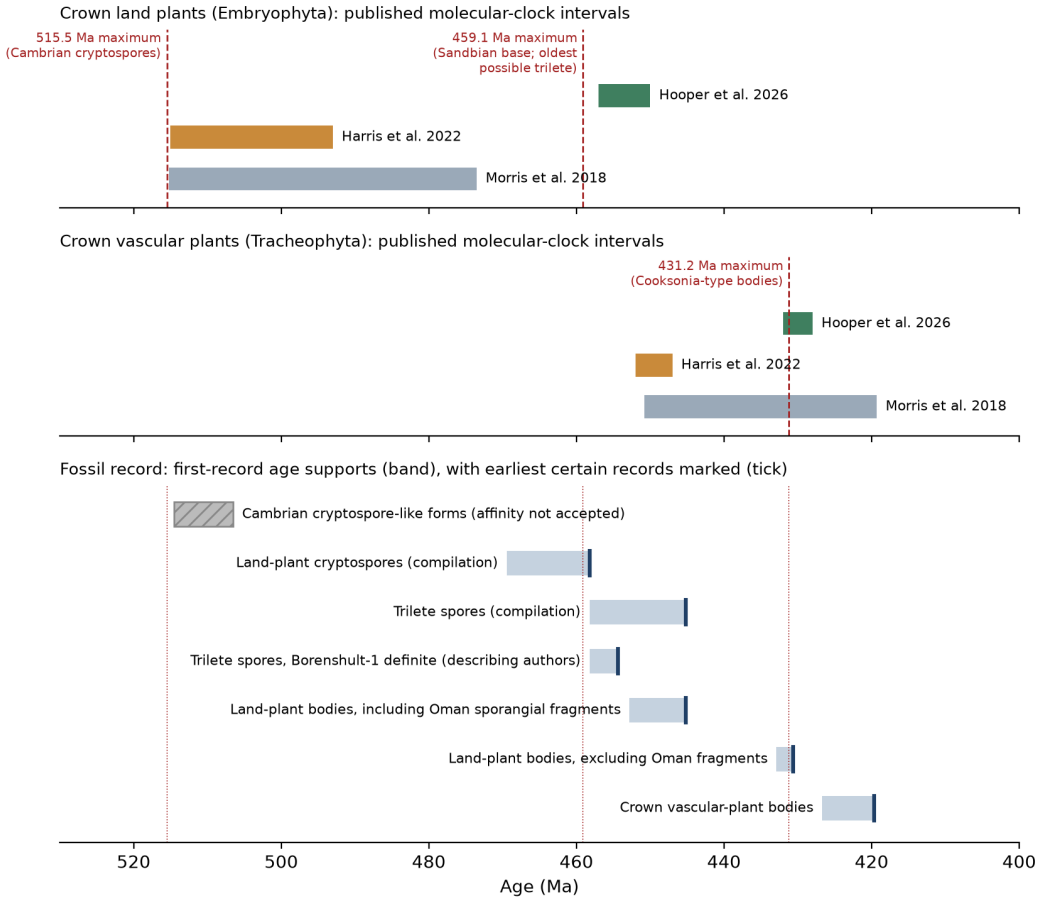


Figure 1: The problem. Top two panels: published crown-age intervals from three molecular-clock studies (Morris et al. 2018b; Harris et al. 2022; Hooper et al. 2026), with the three calibration maxima examined here (dashed). The bars do not share one statistical meaning: Morris et al.'s are envelopes of 95% highest-posterior-density intervals across seven topologies; the others are the intervals each study reports. Bottom: first-record age supports by source and reading. Bands span the earliest possible and earliest certain occurrence (tick) of each category; endpoints may come from different finds, and bands do not show a continuous record. "Compilation" bars are the strict-set first appearances of the project spore compilation. The Borenshult-1 bar is the describing authors' definite-trilete reading (Rubinstein and Vajda 2019): Sandbian, below the 454.41 Ma bentonite; it is shown beside the compilation, not substituted for it. Body bars are the reviewed locality units of this study: land-plant bodies including the Katian Oman sporangial fragments (canonical) (Wellman, Osterloff, and Mohiuddin 2003) and, for comparison, excluding them. The 459.1 Ma maximum is the GTS2012 Sandbian base plus its uncertainty. The hatched bar marks Cambrian Stage 4 cryptospore-like forms, whose land-plant affinity none of the sources examined accepts.

Fig. 2.

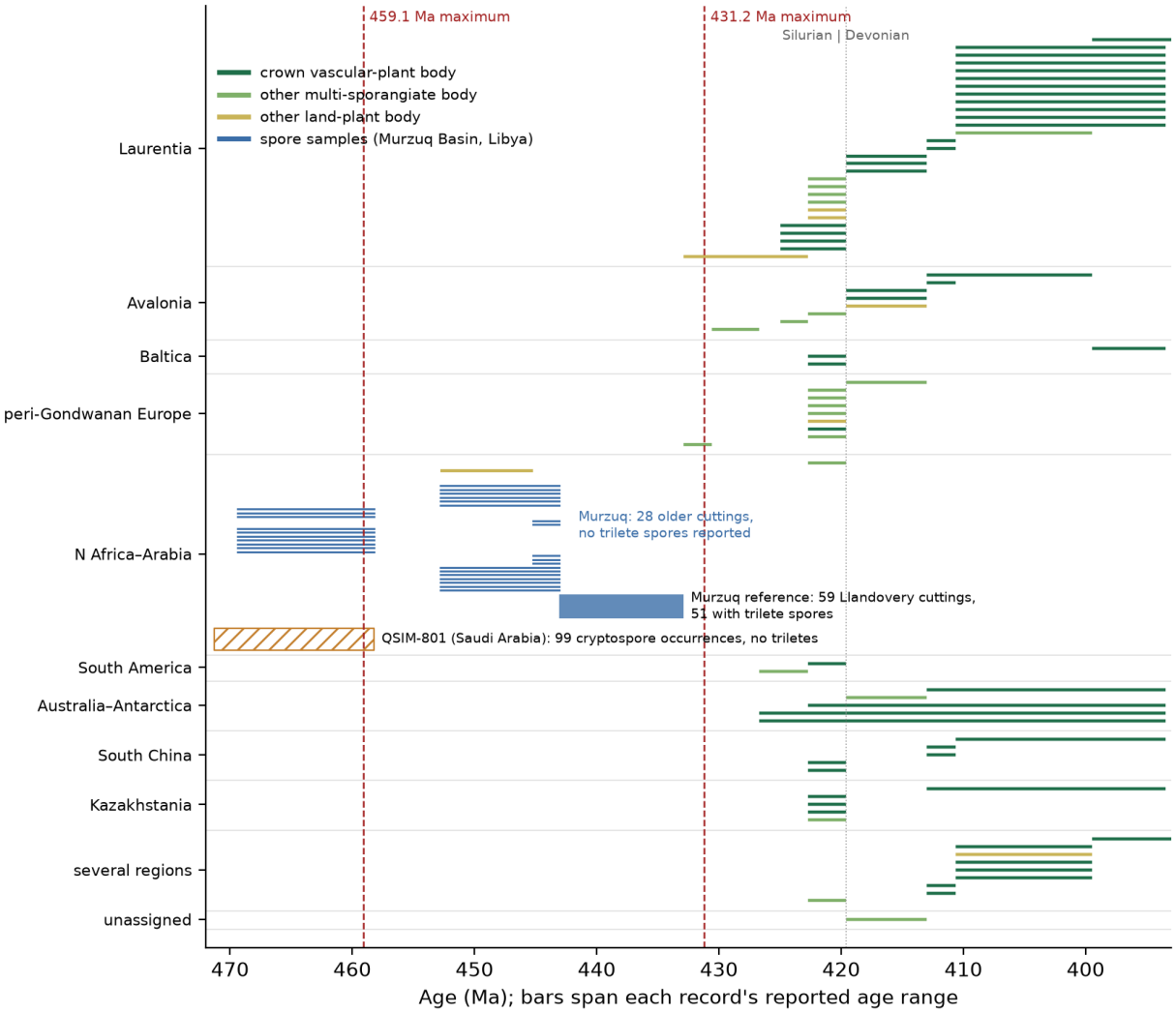


Figure 2: Where the tested evidence comes from. Body-fossil locality units (main reading, canonical inventory including the Oman fragments) by region; bars span reported age ranges, coloured by the most specific category each unit supports. Blue: Murzuq Basin cuttings used in the spore test (older cuttings without trilete reports, and the Llandovery reference block). Hatched: QSIM-801 cryptospore samples (full-stage enclosure; the source assigns them more finely, to the late Dapingian–early Darriwilian). Regions are approximate modern country or state assignments, not a palaeogeographic reconstruction. Spore records outside these two frames are not shown.

Fig. 3.

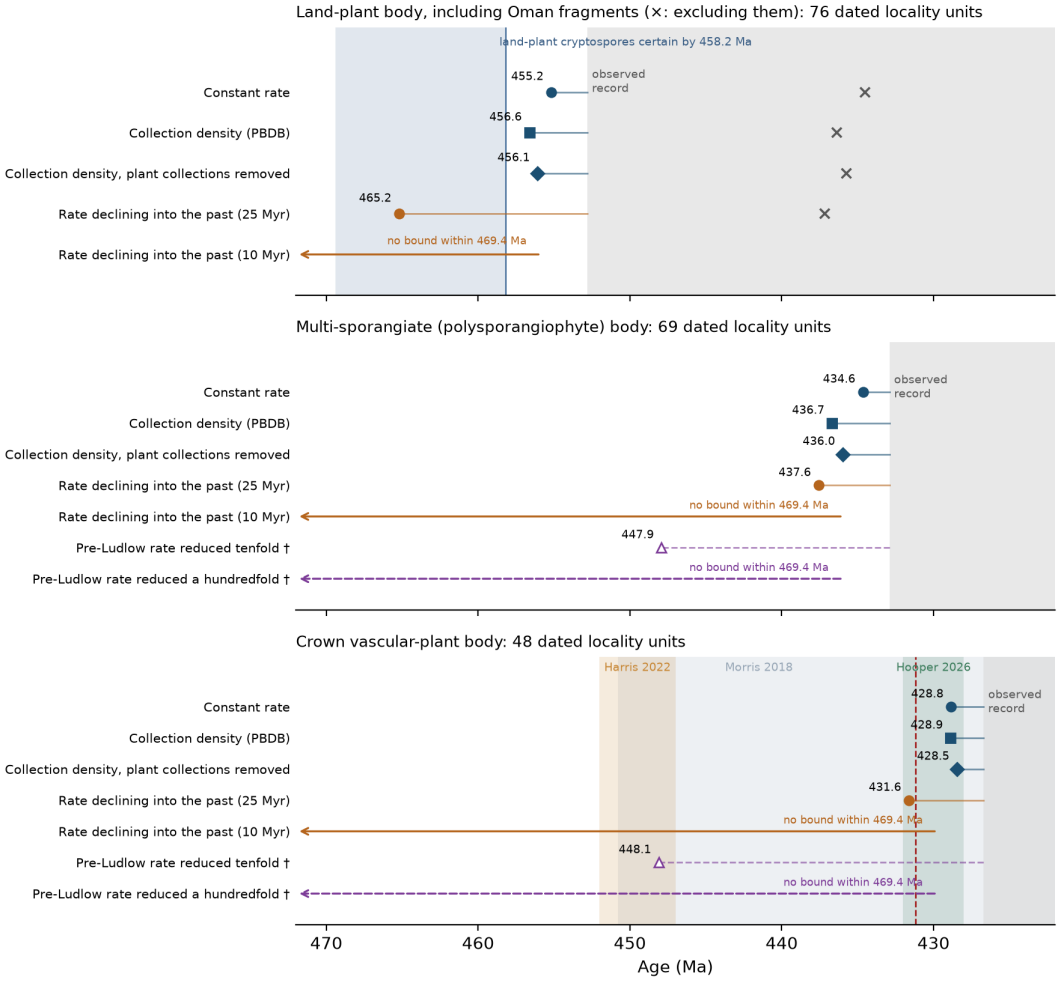


Figure 3: Older bounds on the start of the recognisable body record. Each row is one declared history of the rate of recorded, identifiable occurrences; points are conservative 95% older bounds conditional on that history, the adopted timescale and independent locality–age units (main reading). Grey: span of the observed units. Top: land-plant bodies including the Katian Oman sporangial fragments; × marks the bounds when they are excluded. These bounds are sensitivity arithmetic, not dates at which bodies became recoverable. Open triangles (†) are illustrative scenarios, chosen after the analysis, in which the rate before 426.7 Ma is reduced; for the crown-member category they leave the fit to the observed record unchanged. Top: blue shading marks land-plant cryptospores, possible from 469.4 Ma and certain by 458.2 Ma. Bottom: shaded bands are published crown vascular-plant intervals and the dashed line is the 431.2 Ma maximum; they are comparators, not validation targets. Arrows: no bound within the 469.4 Ma limit of the declared histories.

Fig. 4.

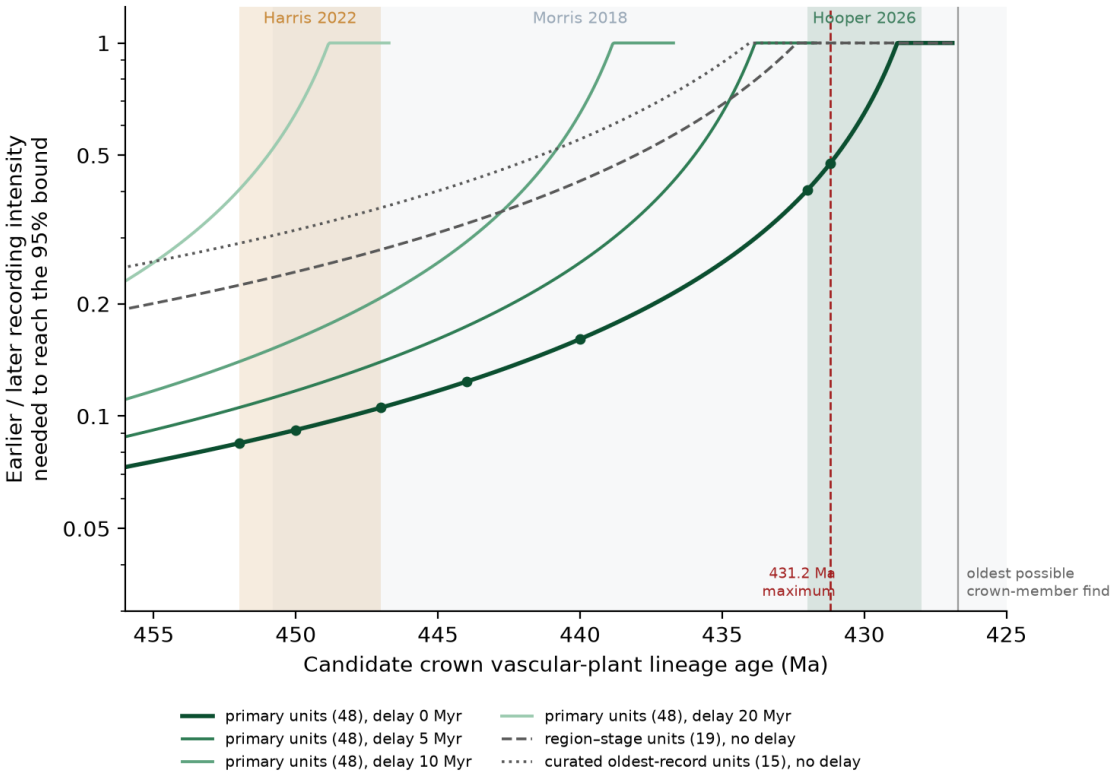


Figure 4: What an older crown would require. For each candidate crown vascular-plant lineage age, the ratio of earlier to later recording intensity at which the conservative older bound on the recognisable record reaches the candidate age minus the assumed diagnostic delay, using the 0.05 comparison rule and a retrospective step at the oldest possible crown-member find (426.7 Ma). Green curves: primary locality–age units, with an assumed delay of 0, 5, 10 or 20 Myr between lineage origin and the start of the recognisable body record; points are the tabulated zero-delay contours. Grey curves: coarser unit treatments, no delay. Ratios at or above 1 are displayed at 1 (no reduction needed); curves are omitted where the candidate age minus delay is no older than 426.7 Ma, where this contour is undefined. Shaded bands are published crown vascular-plant intervals and the dashed line is the 431.2 Ma maximum. The step was placed after seeing the data, so the curves are compatibility requirements of an illustrative model, not fitted rates, estimates of fossilisation probability or a confidence region; the observed records fit equally well at every point on a curve. Closed form and table: Supplement S5.4, Table S9b.